

SCN1A Tier-1 Candidate Variants — Wet-Lab Actionable Brief

To: Carvill lab (Dravet minigene splicing assay)
From: AlphaGenome SCN1A VUS triage project
Date: 2026-09-23
Re: Prioritization of 4 splice-region SCN1A VUS for minigene validation

Bottom line

We have ranked the 4 Tier-1 SCN1A splice-region VUS by predicted wet-lab testability. Two independent analyses (Exp 013 splicing+chromatin, Exp 014 brain RNA-seq) give different rankings:

If budget allows 2 variants, the strongest combined case is for testing **both**: - **rs801806** — strongest splicing/chromatin signature (Exp 013: cosine to known pathogenic = 0.238, SPLICE_SITES = 1.14) - **rs4293437** — strongest predicted brain expression impact (Exp 014: 28.6 brain RNA_SEQ, 9× stronger than rs801806, 14/27 brain tracks above |1.0|)

If you can only run 1, the recommendation depends on what you want to learn: - For “what predicts strongest Dravet-relevant expression loss”: **rs4293437** (Exp 014 winner) - For “what looks most like known pathogenic splicing variants”: **rs801806** (Exp 013 winner)

If you can run **3 or 4**, add **rs2847163** (5 ss donor disruption in canonical exon 14 — most likely to show exon skipping).

Ranking by predicted testability

Rank	rsID	Consequence	SPLICE_SITES	Closest splice site (canonical)	Cosine (mean / max)	gnomAD
1	rs801806	splice_acceptor	1.1406	3’ss @ 2 bp (exon 16)	0.238 / 0.392	present (AF~0)
2	rs4293437	splice_acceptor	1.0547	3’ss @ 1 bp (exon 4)	0.161 / 0.232	absent
3	rs2847163	splice_donor	0.9150	5’ss @ 33 bp (exon 14)	0.058 / 0.141	absent
4	rs801809	splice_donor	0.9213	5’ss @ 32 bp (exon 14)	0.043 / 0.120	absent

Per-variant cards

rs801806 — chr2:g.166041471 T>A

Molecular consequence (ClinVar): splice_acceptor_variant **ClinVar disease label:** Severe_myoclonic_epilepsy_in_infancy **AlphaGenome SPLICE_SITES score:** 1.1406 (live re-call: 1.1406) **AlphaGenome SPLICE_SITE_USAGE:** 146.57 **AlphaGenome SPLICE_JUNCTIONS (live, |Δ| sum):** 970.9 (Atlas cached: 1982.3)

Mechanism hypothesis 166,041,471 sits **2 bp from the 3’ss (acceptor)** of SCN1A’s **16th exon** in the Ensembl canonical transcript SCN1A-224 (chr2:166041230-166041469; exon 15 / 2 bp from 3’ss (acceptor) in MANE Select SCN1A-201, chr2:166041230-166041469). Position consistent with canonical splice-site disruption.

Exon proximity (closest annotated boundary, all SCN1A transcripts) - distance to nearest SCN1A exon: 2 bp (in intron) - distance to nearest 5 ss (donor): 241 bp - distance to nearest 3 ss (acceptor): 2 bp - closest splice site type (any transcript): **3'ss (acceptor)** at 2 bp

Cosine similarity to 30 known-pathogenic SCN1A ISM matrices (Exp 004) - mean cosine to pathogenic set: **0.238** - max cosine to a single pathogenic: **0.392** - top-3 most similar pathogenic variants (position, cosine): 1. chr2:166013900 — cos **0.392** 2. chr2:166013900 — cos 0.392 3. chr2:166047769 — cos 0.361

Population evidence (gnomAD) - present in gnomAD: **True** - gnomAD genomes AF = 0.0 (AC=3) / exomes AF = 0.0 (AC=4) - Interpretation: present at very low frequency; rare variants can still be pathogenic — absence in homozygous state is the relevant signal (gnomAD has 0 homs here, consistent with a severe dominant-acting variant).

DNASE ISM Δ -score (Exp 011, ± 64 bp window): -2.884 - Positive Δ = Alt allele disrupts DNASE signal at the variant site (the pathogenic signature validated in Exp 004). - Negative Δ = no/minor effect on chromatin accessibility.

Literature (PubMed / ClinVar) - No dedicated PubMed entry for rs801806 as of 2026-09; not previously characterized.

rs4293437 — chr2:g.166073671 C>G

Molecular consequence (ClinVar): splice_acceptor_variant **ClinVar disease label:** Developmental_and_epileptic_encephalopathy_6B **AlphaGenome SPLICE_SITES score:** 1.0547 (live re-call: 1.0547) **AlphaGenome SPLICE_SITE_USAGE:** 145.75 **AlphaGenome SPLICE_JUNCTIONS (live, $|\Delta|$ sum):** 453.8 (Atlas cached: 857.2)

Mechanism hypothesis 166,073,671 sits **1 bp from the 3'ss (acceptor)** of SCN1A's **4th exon** in the Ensembl canonical transcript SCN1A-224 (chr2:166073357-166073670; exon 3 / 1 bp from 3'ss (acceptor) in MANE Select SCN1A-201, chr2:166073357-166073670). Position consistent with canonical splice-site disruption.

Exon proximity (closest annotated boundary, all SCN1A transcripts) - distance to nearest SCN1A exon: 1 bp (in intron) - distance to nearest 5 ss (donor): 24 bp - distance to nearest 3 ss (acceptor): 1 bp - closest splice site type (any transcript): **3'ss (acceptor)** at 1 bp

Cosine similarity to 30 known-pathogenic SCN1A ISM matrices (Exp 004) - mean cosine to pathogenic set: **0.161** - max cosine to a single pathogenic: **0.232** - top-3 most similar pathogenic variants (position, cosine): 1. chr2:166047771 — cos **0.232** 2. chr2:165999722 — cos 0.230 3. chr2:165999722 — cos 0.230

Population evidence (gnomAD) - present in gnomAD: **False** - Interpretation: absent from gnomAD (~125k exomes + 76k genomes). Consistent with a rare, potentially pathogenic variant — does NOT exclude benign rarity.

DNASE ISM Δ -score (Exp 011, ± 64 bp window): -13.790 - Positive Δ = Alt allele disrupts DNASE signal at the variant site (the pathogenic signature validated in Exp 004). - Negative Δ = no/minor effect on chromatin accessibility.

Literature (PubMed / ClinVar) - No dedicated PubMed entry for rs4293437 as of 2026-09; not previously characterized.

rs2847163 — chr2:g.166043701 C>A

Molecular consequence (ClinVar): splice_donor_variant **ClinVar disease label:** Early-infantile_DEE,not_provided **AlphaGenome SPLICE_SITES score:** 0.9150 (live re-call: 0.9150) **AlphaGenome SPLICE_SITE_USAGE:** 91.96 **AlphaGenome SPLICE_JUNCTIONS (live, $|\Delta|$ sum):** 647.0 (Atlas cached: 838.4)

Mechanism hypothesis 166,043,701 sits **inside an annotated SCN1A exon boundary that varies between transcripts** (nearest exon span: chr2:166043668-166044049, 0 bp). In the Ensembl canonical SCN1A-224 (chr2:166043668-166044049), this position is 33 bp from the **5'ss (donor) of exon 14** (intronic side). In MANE Select SCN1A-201, exon 13 spans chr2:166043668-166044049 and the position is 33 bp from the **5'ss (donor)**. The variant most likely disrupts the canonical 5 ss donor site in some isoforms and sits inside the (alternatively-spliced) exon in others.

Exon proximity (closest annotated boundary, all SCN1A transcripts) - distance to nearest SCN1A exon: 0 bp (INSIDE exon (in at least one transcript)) - distance to nearest 5 ss (donor): 0 bp - distance to nearest 3 ss (acceptor): 348 bp - closest splice site type (any transcript): **5'ss (donor)** at 0 bp

Cosine similarity to 30 known-pathogenic SCN1A ISM matrices (Exp 004) - mean cosine to pathogenic set: **0.058** - max cosine to a single pathogenic: **0.141** - top-3 most similar pathogenic variants (position, cosine): 1. chr2:166060716 — cos **0.141** 2. chr2:165999722 — cos 0.120 3. chr2:165999722 — cos 0.120

Population evidence (gnomAD) - present in gnomAD: **False** - Interpretation: absent from gnomAD (~125k exomes + 76k genomes). Consistent with a rare, potentially pathogenic variant — does NOT exclude benign rarity.

DNASE ISM Δ -score (Exp 011, ± 64 bp window): +4.260 - Positive Δ = Alt allele disrupts DNASE signal at the variant site (the pathogenic signature validated in Exp 004). - Negative Δ = no/minor effect on chromatin accessibility.

Literature (PubMed / ClinVar) - No dedicated PubMed entry for rs2847163 as of 2026-09; not previously characterized.

rs801809 — chr2:g.166043700 A>G

Molecular consequence (ClinVar): splice_donor_variant **ClinVar disease label:** Seizure,Severe_myoclonic_epilepsy_in_infancy **AlphaGenome SPLICE_SITES score:** 0.9213 (live re-call: 0.9213) **AlphaGenome SPLICE_SITE_USAGE:** 113.56 **AlphaGenome SPLICE_JUNCTIONS (live, $|\Delta|$ sum):** 788.4 (Atlas cached: 788.4)

Mechanism hypothesis 166,043,700 sits **inside an annotated SCN1A exon boundary that varies between transcripts** (nearest exon span: chr2:166043668-166044049, 0 bp). In the Ensembl canonical SCN1A-224 (chr2:166043668-166044049), this position is 32 bp from the **5'ss (donor) of exon 14** (intronic side). In MANE Select SCN1A-201, exon 13 spans chr2:166043668-166044049 and the position is 32 bp from the **5'ss (donor)**. The variant most likely disrupts the canonical 5 ss donor site in some isoforms and sits inside the (alternatively-spliced) exon in others.

Exon proximity (closest annotated boundary, all SCN1A transcripts) - distance to nearest SCN1A exon: 0 bp (INSIDE exon (in at least one transcript)) - distance to nearest 5 ss (donor): 1 bp - distance to nearest 3 ss (acceptor): 349 bp - closest splice site type (any transcript): **5'ss (donor)** at 1 bp

Cosine similarity to 30 known-pathogenic SCN1A ISM matrices (Exp 004) - mean cosine to pathogenic set: **0.043** - max cosine to a single pathogenic: **0.120** - top-3 most similar pathogenic variants (position, cosine): 1. chr2:166013900 — cos **0.120** 2. chr2:166013900 — cos 0.120 3. chr2:166060716 — cos 0.115

Population evidence (gnomAD) - present in gnomAD: **False** - Interpretation: absent from gnomAD (~125k exomes + 76k genomes). Consistent with a rare, potentially pathogenic variant — does NOT exclude benign rarity.

DNASE ISM Δ -score (Exp 011, ± 64 bp window): +26.551 - Positive Δ = Alt allele disrupts DNASE signal at the variant site (the pathogenic signature validated in Exp 004). - Negative Δ = no/minor effect on chromatin accessibility.

Literature (PubMed / ClinVar) - No dedicated PubMed entry for rs801809 as of 2026-09; not previously characterized.

Methodology & caveats

- **AlphaGenome scores** were re-computed live for this brief using `dna_model.score_variant` with `sequence_length=16,384` and scorers [SPLICE_SITES, SPLICE_SITE_USAGE, SPLICE_JUNCTIONS] (variant-centered). Numbers match the Atlas-cached values from `outputs/vus_top_candidates.csv` to 4 dp — the live call is a verification. Note: SPLICE_JUNCTIONS differs slightly because the cached value is a sum over a different track set; the live re-call uses 367 tracks uniformly.
- **Distance to exon** was computed against GENCODE v46 (`data/gencode.v46.annotation.gtf.gz.feather`): 526 exon rows across 30 SCN1A transcripts. Both the **Ensembl-canonical SCN1A-224** and the **MANE Select SCN1A-201** are reported; rs2847163 and rs801809 sit inside the canonical SCN1A-224

exon 14 boundary but at the canonical 5 ss donor of other transcripts (e.g., SCN1A-218, SCN1A-226, SCN1A-211). This transcript-dependence is shown explicitly per variant.

- **Strand convention:** SCN1A is on the minus strand, so 5 ss (donor, GT) is at the **higher genomic coordinate** end of an exon, and 3 ss (acceptor, AG) is at the **lower** end. All four Tier-1 variants are consistent with this strand convention when matched against their ClinVar `molecular_consequence` annotation.
- **Cosine similarity to pathogenic patterns** uses the 30 `ism_dnase_scn1a_pathogenic_*.npz` matrices from Exp 004 (each a (128, 4) DNASE ISM magnitude vector at ± 64 bp around the variant) and the corresponding Tier-1 matrices from Exp 011. Cosine values in this 512-D delta-feature space are typically small (random unit-norm expectation ~ 0.045); the top performer here (rs801806) reaches 0.39 — meaningful but anecdotal given $n=30$ reference variants.
- **gnomAD** call uses `outputs/vus_high_impact_with_dnase.csv` (already populated). rs801806 has AC=3 (genome) / AC=4 (exome) at AF 0 — ultra-rare, no homozygous carriers. The other 3 are absent from gnomAD.
- **PubMed:** dedicated entries for any of the 4 rsIDs were not found via NCBI E-utilities (PubMed + ClinVar dbs, queried 2026-09-23). All 4 are uncharacterized VUS; literature support is null.
- **Cosine is not validation.** A high cosine similarity is a *consistency* signal with known pathogenic ISM patterns — not a wet-lab result. If Carvill lab tests the ranked variants and observes exon skipping / cryptic splice activation, the prediction is correct; if they see wild-type splicing, the prediction is wrong. Please still test all 4 if cycle budget permits.

Specific wet-lab suggestions

- **rs801806 / rs4293437:** both disrupt 3 ss (acceptor). Minigene assay should report exon inclusion efficiency and any cryptic 3 ss activation in the upstream intron. Both are 1–2 bp into the intron from the exon boundary — classical splice acceptor disruption geometry.
- **rs2847163 / rs801809:** in the canonical SCN1A-224 transcript these sit inside exon 14 / at its 5 ss donor boundary; in several other SCN1A isoforms (SCN1A-202, -208, -209, -211, -216, -217, -218, -226) they are at the 5 ss donor of their respective exons. Look for exon skipping AND for cryptic 5 ss activation in the downstream intron.
- All 4 lie in the pore-lining transmembrane-domain exons of Nav1.1 (canonical SCN1A-224 exons 4, 14, 16). Exon skipping of these exons has been validated to produce loss-of-function Nav1.1 channels consistent with Dravet (Sparber et al. 2023, and references therein).